The 4th Down Equation: Analyzing the Interaction Between Offensive EPA and Defensive Pass-Prevention Tendencies

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Overview

We investigated whether a defense's pass-prevention quality, measured by average depth of target (aDOT) allowed and deep pass EPA allowed, affects offensive outcomes on 4th down conversion attempts. Initially, we looked at the 2024 and 2025 NFL regular season data on 4th downs. Our target was EPA prediction, and we ran a general linear regression, split linear regressions of pass vs. run, and linear regression with an interaction term between aDOT allowed and pass play. All our models failed to reject the null in that nearly all of our features were not statistically significant, thus our chosen defensive metrics have no bearing on the EPA of a 4th down attempt.

We then pivoted to figuring out possible reasons why the defense was inconclusive to the expected addition of points. For this we needed more data, because of our initial dataset, only about 20% (1,747 plays) of recorded 4th downs were conversion attempts. Five NFL regular seasons (2020–2024) of nflverse PBP (play by play) data were used, covering 3,800+ conversion attempt plays before filtering. 'Desperate' plays (win probability below 5%, or ≤10 seconds remaining, or ≤60 seconds remaining while ≥40 yards from the end zone with Win Probability < 15%) were excluded to isolate actual strategic decisions, removing 1,026 plays (about 27% of all conversion attempts). Complementary analyses were conducted using Defensive Tiers as well as Run vs. Pass Conversions. The central finding across all analyses is a consistent null: defensive pass prevention quality has no detectable effect on 4th-down outcomes. The only robust predictor of success is yards to go, and the run game continues to outperform the pass substantially.

Initial Investigation

We attempted to predict EPA using several features directly from the dataset as well as aggregated defensive metrics generated for each time and seasonally. Unfortunately, our models yielded no statistically significant predictors of EPA when it came to our defensive metrics. In fact, with near 0 R^2 values, we concluded there must be a flaw with our approach to the question.

OLS Regression

	Coef	Std err	t	P > |t|	[0.025	0.975]
Intercept	2.0317	1.247	1.630	0.103	-0.414	4.477
ydstogo	-0.0639	0.024	-2.628	0.009	-0.112	-0.016
yardline_100	0.0042	0.004	1.028	0.304	-0.004	0.012
adot_allowed	-0.1839	0.159	-1.154	0.249	-0.496	0.129
pass_play	-0.4628	0.190	-2.440	0.015	-0.835	-0.091

R^2: 0.018

OLS with Interaction Term

	Coef	Std err	t	P > |t|	[0.025	0.975]
Intercept	1.1958	1.826	0.655	0.513	-2.385	4.777
ydstogo	-0.0781	0.020	-3.871	0.000	-0.118	-0.039
yardline_100	0.0041	0.003	1.184	0.237	-0.003	0.011
pass_play	0.0520	2.250	0.023	0.982	-4.362	4.466
adot_allowed	-0.0739	0.236	-0.314	0.754	-0.536	0.388
pass_x_adot	-0.0557	0.290	-0.192	0.848	-0.625	0.513

R^2: 0.020

Interaction between aDOT allowed and pass play yielded a p-value of 0.848, this means there is no statistically significant evidence that aDOT allowed moderates the EPA effect of passing on 4th down, given the model and sample.

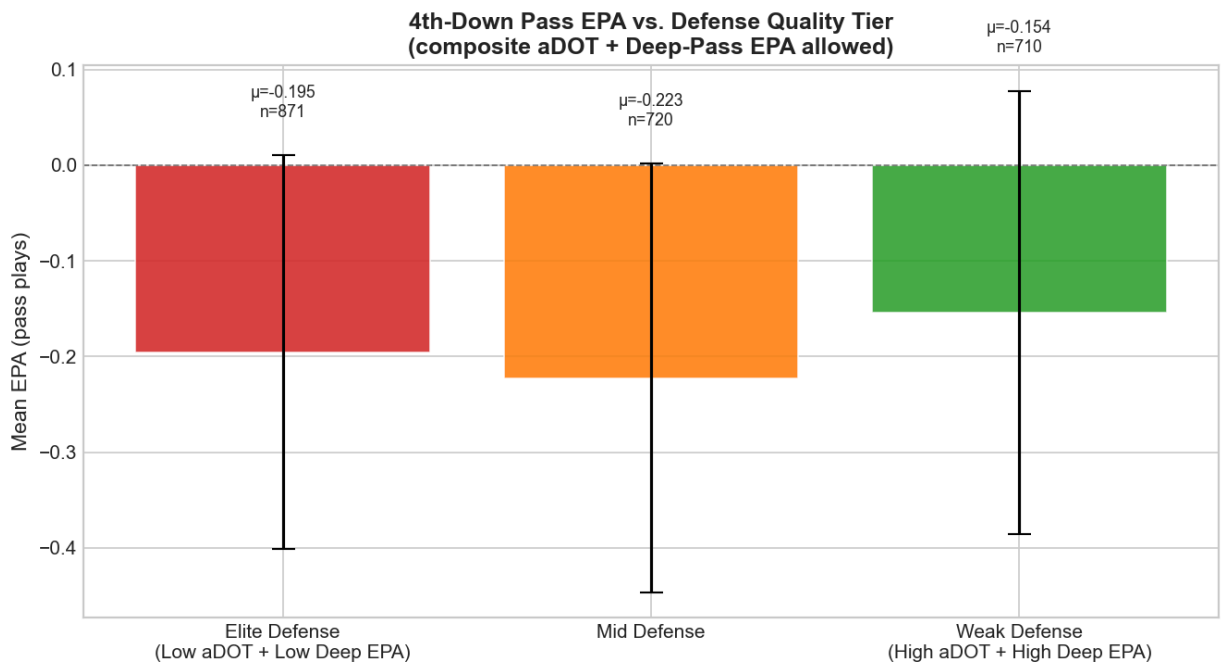
The initial framing of the question produced uniformly inconclusive results. 4th down conversion attempts are an extremely rare occurrence in the context of an entire NFL game, even with the large uptick in 4th down attempts in the 2020s, only about 2-3 occur per game. Working with these plays is extremely hard due to the extremely high variance in context and low sample size. Additionally, the scope of our metrics do not mesh well together: aDOT allowed is a seasonal aggregate and we attempted linking it to individual plays, and this clearly resulted in lots of noise. EPA is already not a very clear metric, and restricting it to 4th downs, which are some of the most unpredictable and highest stake plays, makes our results extremely difficult to capture.

A Different Approach: EPA Through Defensive Tiers

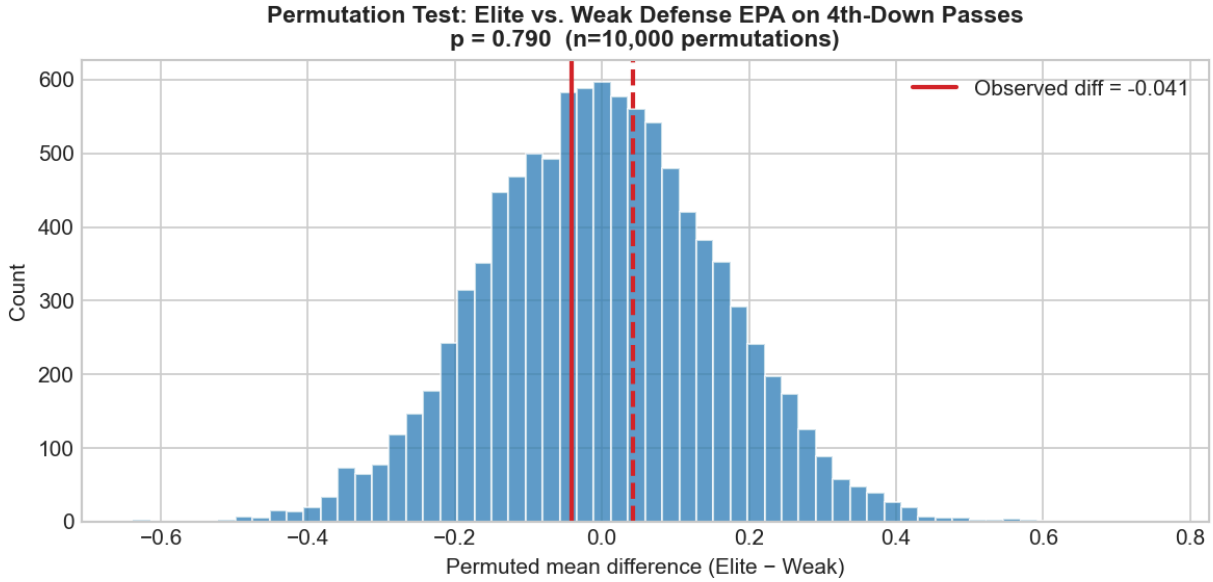
Defenses were stratified into three annual tiers (Elite, Mid, Weak) using a composite z-score of aDOT allowed and deep pass EPA allowed, computed from all season passing attempts to ensure adequate sample size (700-1,300 pass plays per team and season, vs. only 15-30 4th down pass plays). The mean EPA on 4th down pass plays against each tier:

Defensive Tier	N	Mean EPA	95% Bootstrap CI
Elite Defense	871	-0.195	[-0.399, 0.009]
Mid Defense	720	-0.223	[-0.459, 0.002]
Weak Defense	710	-0.154	[-0.388, 0.078]

The observed gap between Elite and Weak defenses is only -0.041 EPA which is essentially nothing, considering the play-level standard deviation exceeds 3.0. All three bootstrap confidence intervals overlap substantially and each includes zero.



A permutation test (10,000 iterations) randomly reassigned defensive tier labels across plays and computed the Elite - Weak EPA gap each time. The observed -0.041 difference fell at the 79th percentile of the null distribution ($p = 0.79$), i.e. a gap at least this large appeared 79% of the time purely by chance. So this clearly is in line with the null.

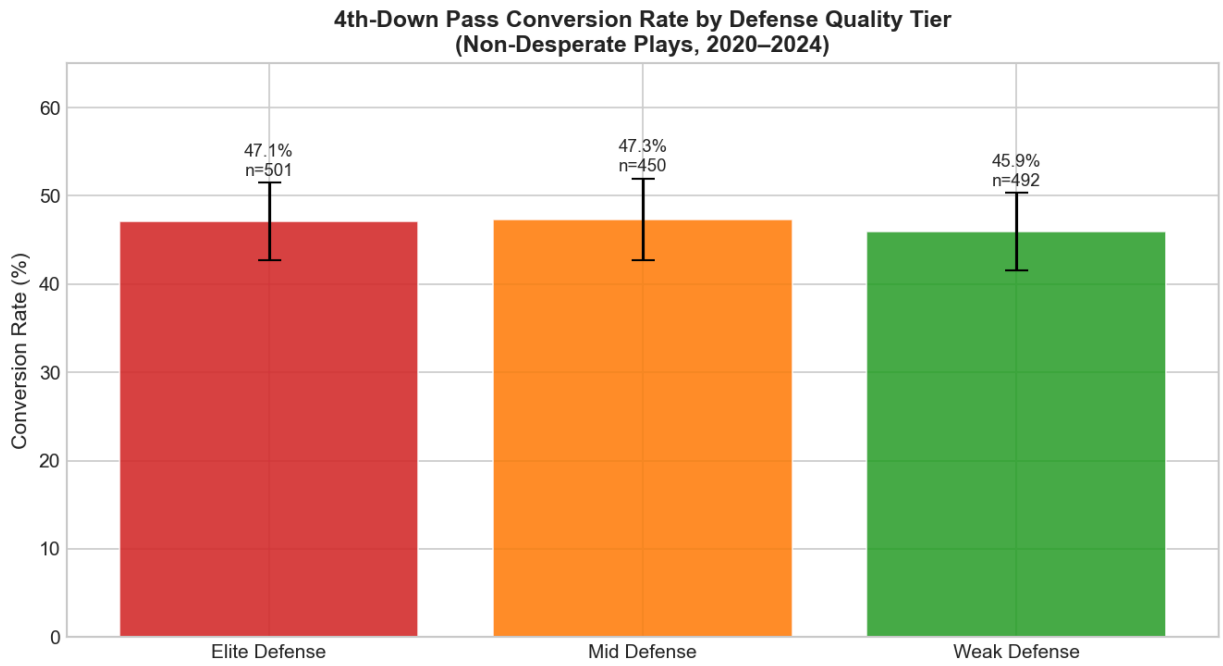


Shifting from EPA to binary conversion (success/failure) removed distributional concerns (outlier long touchdown passes inflate EPA means), but our conclusion did not change.

Conversion Rates

After removing Hail Mary plays (≥40 yard air balls in the final 60 seconds with Win Probability < 15%, in addition to the 'desperate' play filter), 1,443 clean 4th down pass plays remained. Conversion rates by defensive tier were nearly indistinguishable:

Defensive Tier	N	Conversion Rate
Elite Defense	501	47.1%
Mid Defense	450	47.3%
Weak Defense	492	45.9%



A chi-square test across all three tiers returned $p = 0.897$, i.e. there are no statistically significant differences between the groups. A 10,000-iteration permutation test on the Elite - Weak gap returned $p = 0.606$.

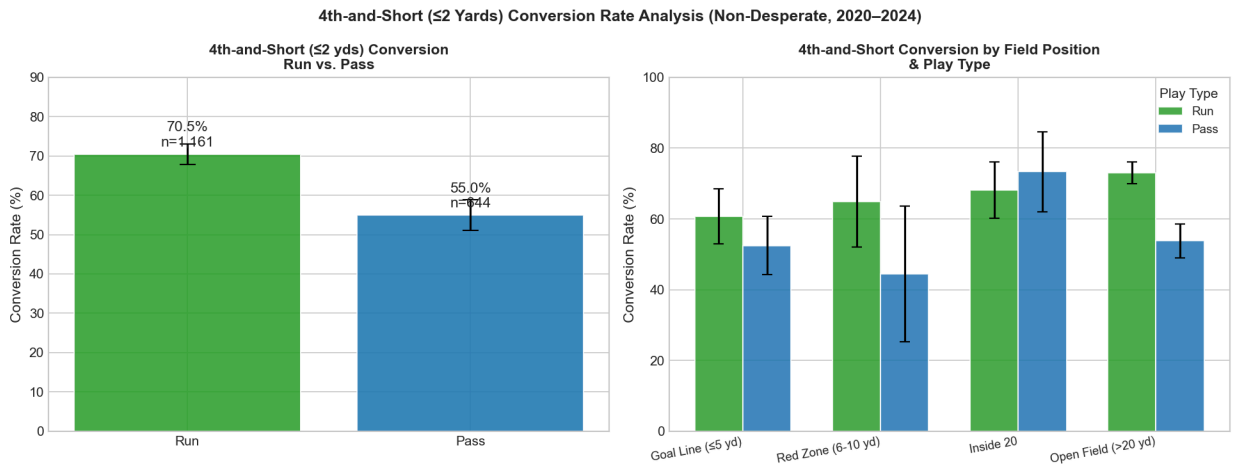
Run vs. Pass for 4th and Short

While the defensive quality null held consistently, the comparison between run and pass on 4th and short (≤2 yards to go) revealed the most practically significant finding.

Across 1,805 4th and short plays (2020–2024), teams ran the ball on 64.3% of attempts (N = 1,161) and passed on 35.7% (N = 644). The outcome gap was large:

Play Type	N	Conversion Rate
Run	1,161	70.7%
Pass	644	51.8%
Difference	-	-18.9 pp

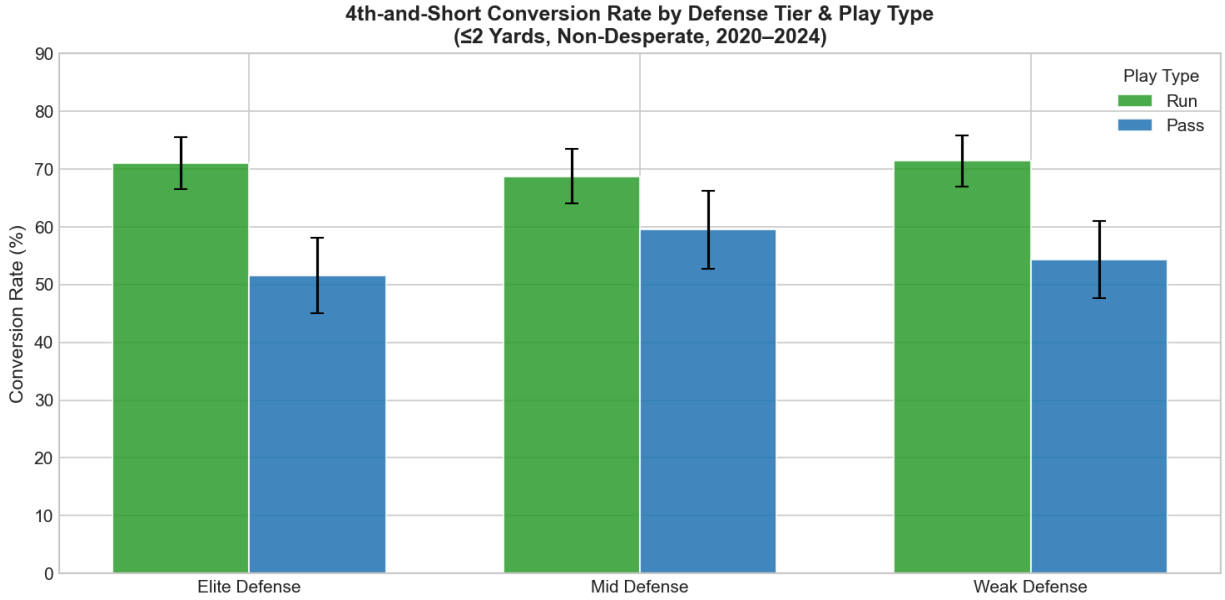
A logistic regression controlling for yards to go, field position, score differential, game clock, and goal-line context found that passing on 4th and short carries an Odds Ratio (OR) = 0.60 ($p < 0.001$), i.e. a pass attempt reduces the odds of converting by 40% relative to a run, even after controlling for situational factors.



Interestingly, the defensive tier null held even within the 4th and short context, where one might expect pass prevention quality to matter more on critical short yardage situations.

Defense	Pass Conv.	Run Conv.
Elite	51.6%	71.0%
Mid	59.5%	68.8%
Weak	54.3%	71.4%

Pass conversion rates across tiers ($p = 0.62$) and run conversion rates ($p = 0.95$) showed no significant variation. aDOT allowed in the short yardage logistic model: OR = 0.95, $p = 0.36$.



Despite the league-wide increase in 4th down attempts, the pass rate on 4th and short has been remarkably stable. Teams are going for it more often, but they are not changing how they go for it on short yardage downs. The run approach still reigns supreme, even as advanced analytics begin to permeate through the NFL.

Conclusion

The robustness of the defensive null is notable. Three different outcome measures (EPA, binary conversion, short-yardage conversion) and different statistical methods (OLS, OLS with Interaction Term, Logistic Regression, Permutation Testing) all returned the same answer. In both investigations, we looked at thousands of conversion attempts spanning multiple seasons, thus we can conclude there is no signal of our chosen defensive metrics on EPA.

One interpretation is that 4th down situations compress variance in a way that 1st/2nd/3rd down analysis does not. Teams are more conservative about when they go for it on 4th down against elite defenses (they might kick more), so the plays that do appear in the sample against elite defenses may already be skewed for favorable yardage or field position. Selection effects of this kind could weaken any true defensive signal.

The run game finding is more straightforward. The 18.9 percentage point gap between run and pass on 4th and short is large, consistent across all five seasons, and holds within every defensive tier. The fact that teams choose to pass 36% of the time on 4th and short in the 2020s, suggests either that coaches are attempting to maintain unpredictability, that specific matchup contexts (heavy pressure risk, goal-line crowding, time left) justify the pass in individual decisions, or simply that old habits persist. Whatever the reason, the aggregate data consistently favor the run on short yardage 4th downs.